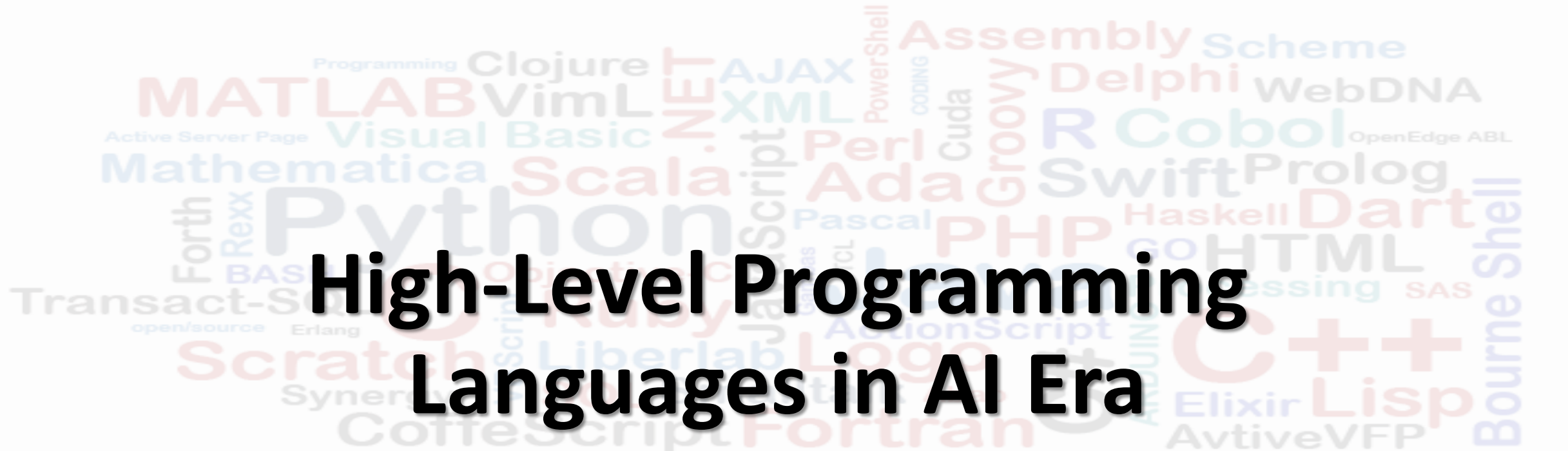
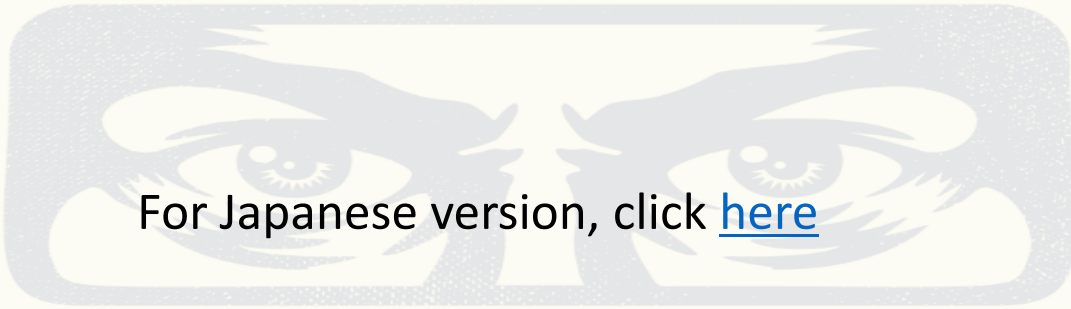


A word cloud featuring various programming languages and technologies. The most prominent words are Python, MATLAB, and Scratch. Other visible words include JavaScript, PHP, C++, Java, C, Fortran, Perl, Ada, Swift, Prolog, Haskell, Dart, R, Cobol, Delphi, Scheme, Assembly, Clojure, Vim, XML, AJAX, PowerShell, Cuda, Groovy, OpenEdge, ABL, WebDNA, Active Server Page, Visual Basic, Mathematica, Scala, Pascal, ActionScript, Elixir, Lisp, AvtiveVFP, Bourne Shell, SAS, Erlang, Synergy, CoffeeScript, Transact-SQL, open/source, Forth, Rexx, and Bash.

High-Level Programming Languages in AI Era

-In Terms of Human Thought and Mind-

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For Japanese version, click [here](#)

Composed By: Hayato Ishida

Updated On: 30 June 2025

About Me

- Accounts
 - GitHub: [@hayat01sh1da](#)
 - X: [@hayat01sh1da](#)
 - LinkedIn: [@hayat01sh1da](#)
 - Speaker Deck: [@hayat01sh1da](#)
 - Docswell: [@hayat01sh1da](#)
- Occupation: Software Engineer
- Things I Am Into
 - Linguistics
 - Singing at Karaoke
 - Listening to Music
 - Watching Movies
 - Cat Feeding



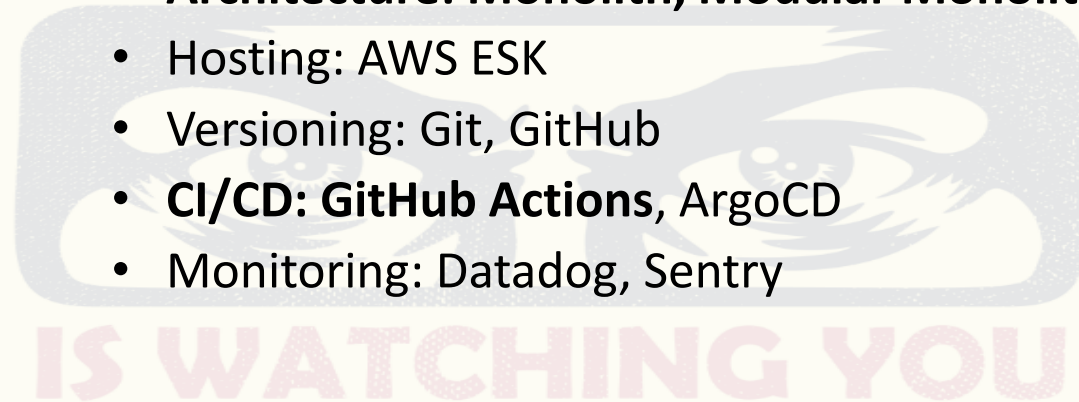
Licences / Certifications

- English
 - TOEIC® Listening & Reading 915: Certified on December 2019
- Engineering
 - Information Security Management: Certified on November 2017
 - Applied Information Technology Engineer: Certified on June 2017
 - Fundamental Information Technology Engineer: Certified on November 2016
 - IT Passport: Certified on April 2016
- Others
 - Abacus 2nd Class: Certified on June 2002
 - Mental Arithmetic 3rd Class: Certified on February 2001



Skill Stack

- Languages
 - Japanese: Native Proficiency
 - English: Full professional Proficiency
- Development
 - **Backend:** Ruby(Ruby on Rails), Python
 - Frontend: TypeScript + React.js, TypeScript +Vue.js
 - **Database:** MySQL, PostgreSQL, MongoDB
 - **Architecture:** Monolith, Modular Monolith
 - Hosting: AWS ESK
 - Versioning: Git, GitHub
 - **CI/CD:** GitHub Actions, ArgoCD
 - Monitoring: Datadog, Sentry



Work Histories

Period	Genre	Position	Jobs
Aug 2021 - Present	SaaS(Educational Service)	Software Engineer	• Backend Development • Frontend Development • CI/CD Maintenance • Weekly Release Manager • Troubleshooting • Documentation • Technical Debt Resolution • Tech Blog Entries • LT Speaker
Feb 2020 - Dec 2020	SaaS(Chatbot Platform)	Backend Engineer	• Backend Development • Documentation • Technical Debt Resolution • Inspection of Alternative API

Work Histories

Period	Genre	Position	Jobs
Jun 2018 - Jan 2020	Contract-basis Developer	Software Engineer	• Backend Development • Frontend Development • Troubleshooting • Quality Assurance • Tech Blog Entries
Apr 2016 - Jan 2018	System Engineering Service	System Engineer	• Corporate Account Admin • Windows Server Maintenance • Security Admin Assistant • Internationalisation / Localisation



International-Exchange Activities

- Activities in University
 - English Linguistics Seminar(Focusing on Mass Media English)
 - International-Exchange Clubs(The 2nd Year)
 - [International-Exchange Programmes conducted by Japan Cabinet Office](#)(2013 - 2016)
 - Japanese Linguistics Course(The Final Year)
- Overseas Life Experience
 - Working Holiday in Australia(April 2014 - March 2015)
 - Language School for 1 month in Sydney
 - Work for 6 Months in [Hamilton Island Resort](#)
 - Volunteering for 1 Month as Assistant Teacher of Japanese Language at [St Ives High School](#) in NSW
- Other Activities
 - Keep Everyday Journal in English (April 2014 - Present)
 - [Sunrise Toastmasters Club](#)(February 2017 - March 2018)
 - [Vital Japan](#)(January 2018 - July 2019, October 2022 - February 2023)
 - Self Learning and Training of English Language
 - Participation Ruby-Related Tech Conferences

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Agenda

1. [Let's Think about...](#)
2. [The Book I would Like to Introduce](#)
3. [High-Level Programming Languages in AI Era](#)
4. [Summary of The Worldview of The Book](#)
5. [Newspeak](#)
6. [Why Nineteen Eighty-Four?](#)
7. [References](#)



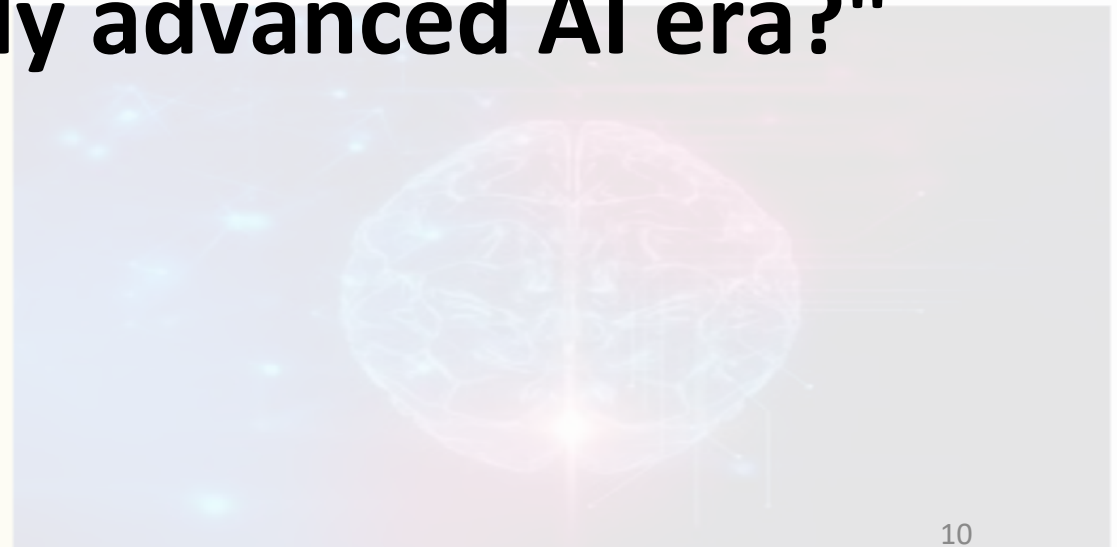


1. Let's Think about...

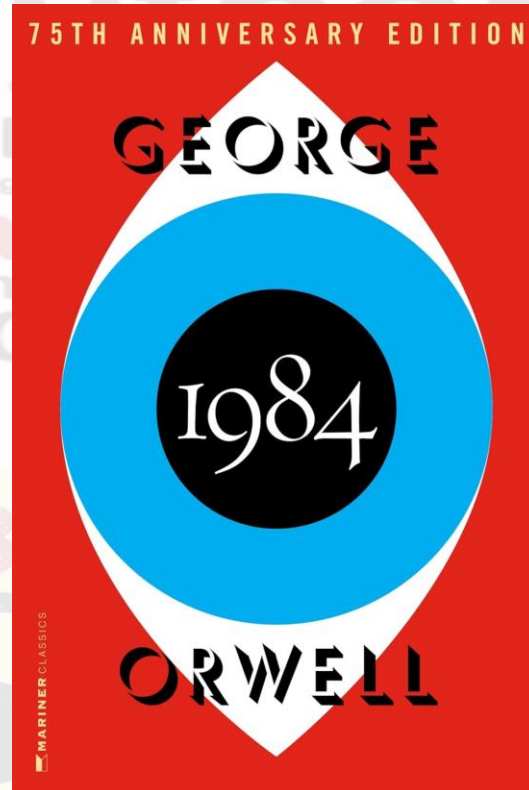
1. Let's Think about...

Even given the reality that AI evolution is still in progress and high-level interface is still in need, the question is...

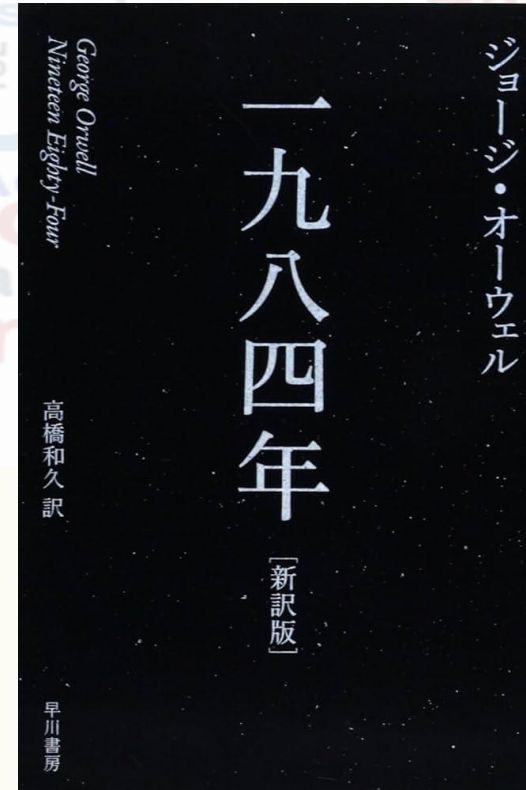
"What do we use high-level programming languages for in this highly advanced AI era?"



2. The Book I would Like to Introduce



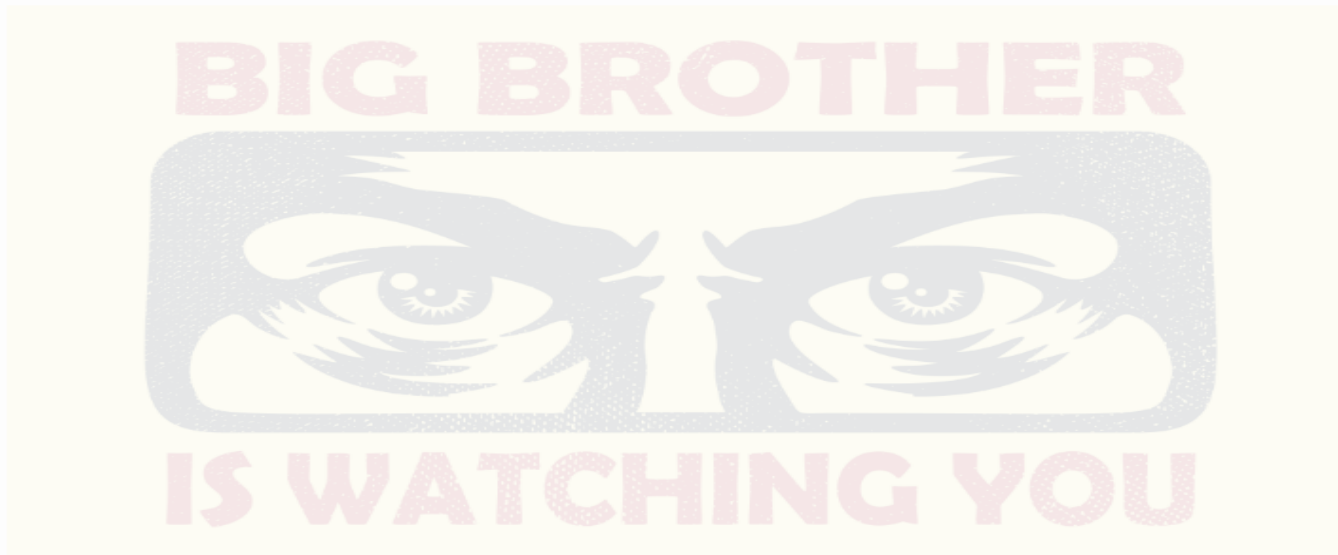
English Edition



Japanese Edition



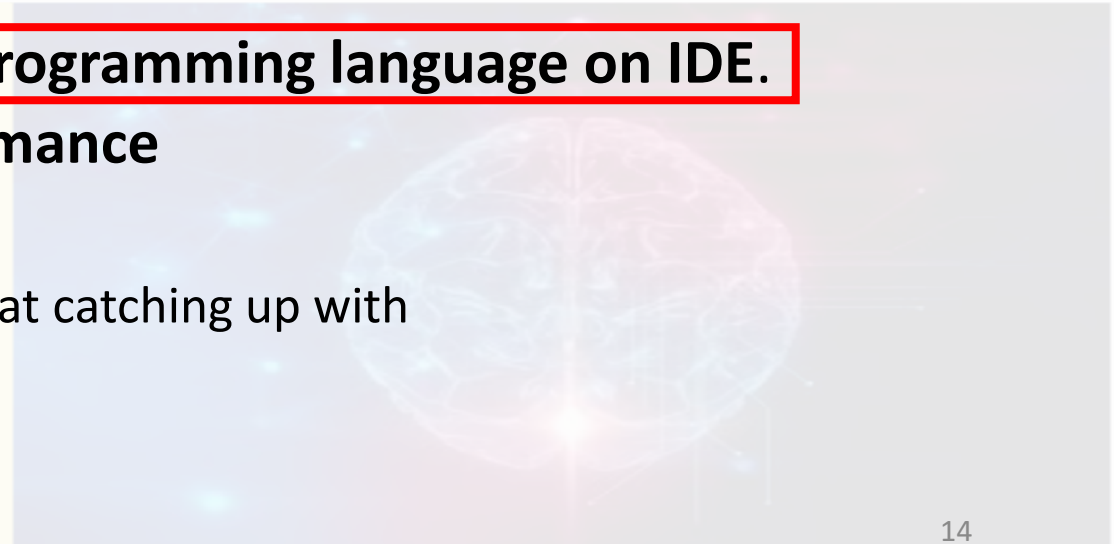
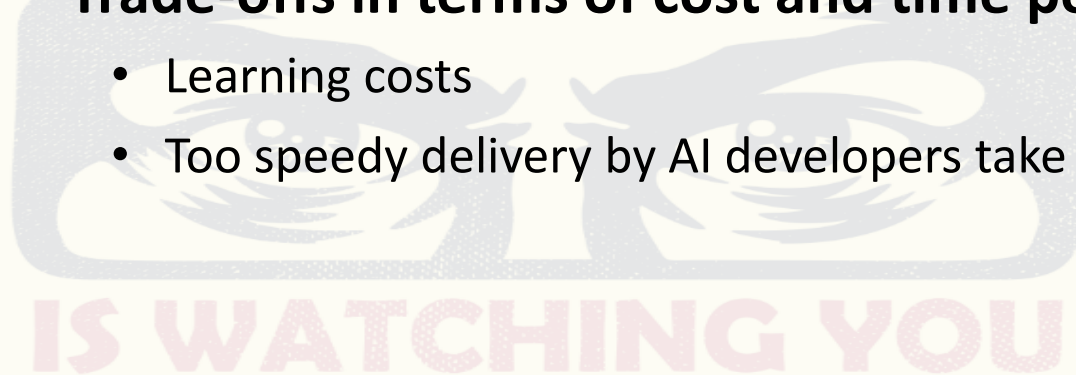
3. High-Level Programming Languages in AI Era



3. High-Level Programming Languages in AI Era

Big Changes to Development Experiences brought by AI Tools

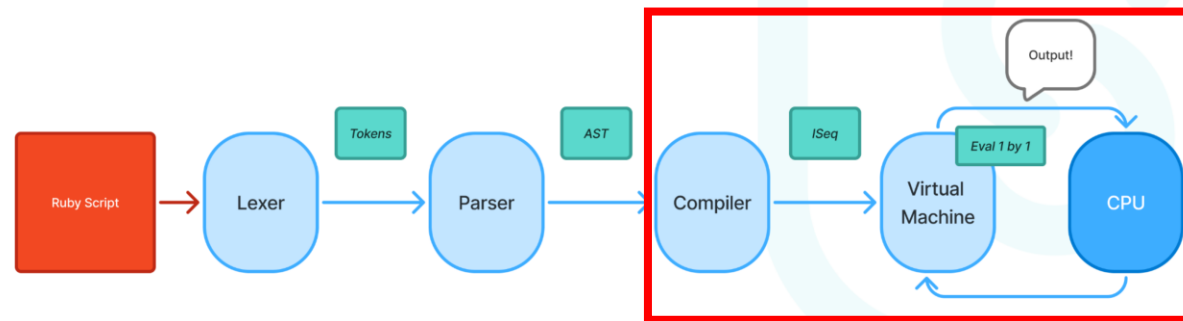
- Rivalry of warlords in progress
 - GitHub Copilot
 - Devin
 - Clause Code
 - Cursor
- **More natural language on a prompt than programming language on IDE.**
- Trade-offs in terms of cost and time performance
 - Learning costs
 - Too speedy delivery by AI developers take pains at catching up with



3. High-Level Programming Languages in AI Era

Were AI to **directly generate expected bytecodes without any bugs**, necessary ecosystem would be only **Compiler and Virtual Machine** in case of Ruby, for instance.

Rubyスクリプトが実行されるプロセス模式図



Necessary Ecosystem in coming 'Complete' AI Era

p.29, Uchio Kondo, *Ruby の中身の予備知識 - RubyKaigi の前に知り網*

3. High-Level Programming Languages in AI Era

What only human developers can but AI cannot in high-level programming language is...

Naming!!

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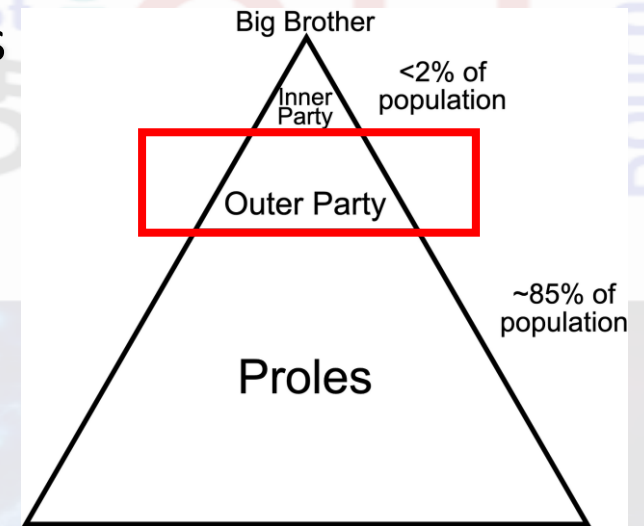
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4. Summary of The Worldview of The Book

4. Summary of The Worldview of The Book

- 3 superstates: Oceania, Eurasia, Eastasia
- Oceanic society: Everything was under control of its government
- Hierarchy: Determined how much controlled the person is
 - Outer Party members: The very target of monitoring
 - Vocabulary: To control thoughts and minds of the Public
 - Thought / Mind: To control political ideologies of the public
 - Political Ideology: To realise [INGSOC\(English Socialism\)](#)
 - etc.
 - Proles: The majority but almost ignored
- The public: Doomed to love [Big Brother](#), who was the very symbol of [INGSOC](#)

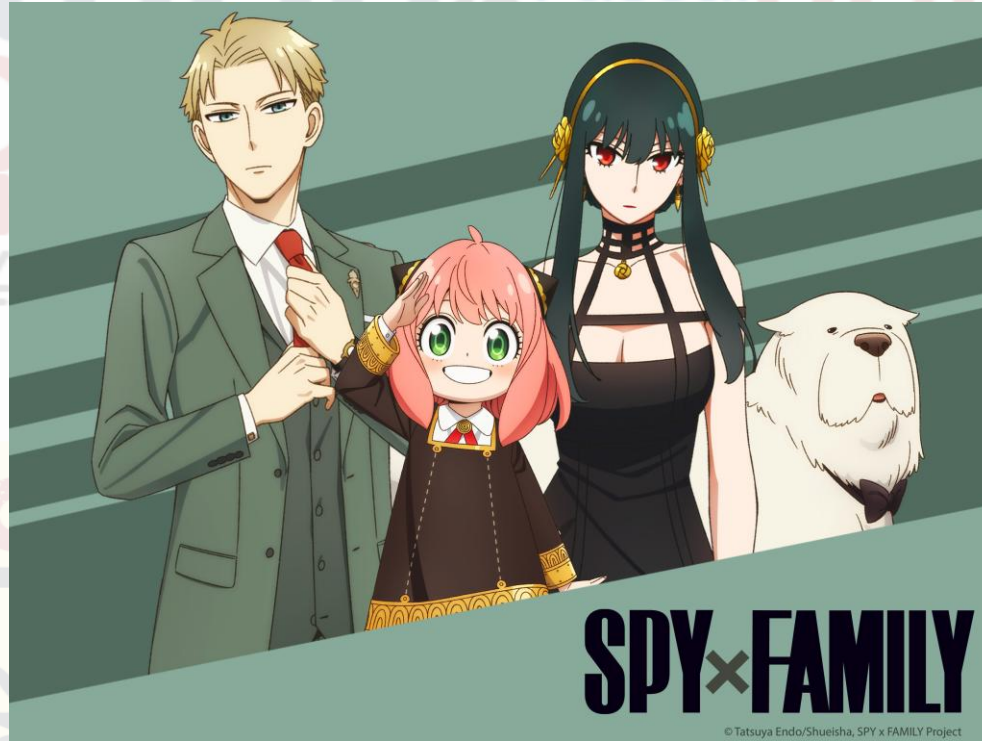


4. Summary of The Worldview of The Book

To roughly compare it to an Anime...



4. Summary of The Worldview of The Book



The worldview of *SPY x FAMILY* might be close a bit.

5. Newspeak

- **Ecosystem:** Another language from **Oldspeak**(the present-day or standard English as of 1984)
- **Commonization: Still in progress**
 - In written context, only *Times*, one of news media, published leading articles in the language
- **Supersedure Plan:** Expected to **exterminate Oldspeak by 2050** in the end
- **Vocabulary: Reduced every year**
 - Evolved in a totally opposite way from the present-day English we know today
- **The biggest aim: The narrower vocabulary, the fewer concepts thinkable**
- **Side effects:** [Economy of Efforts](#) and **loss of awareness of the original concept**
- **Categories:** the A Vocabulary, the B Vocabulary and the C Vocabulary.

5. Newspeak

The A Vocabulary: Everyday Life Uses

- **Vocabulary:** The number was extremely small and rigidly defined
- **No more derivatives:** Among the etymologically connected words
 - thought(noun) / think(verb) → **think**(noun & verb)
 - knife(noun) / cut(verb) → **knife**(noun & verb)
- **Prefixes:** Emphasised or denied the meaning of words
 - **un-** + good → **ungood**(≡ bad)
 - **plus-** + cold → **pluscold**(≡ very cold) > **doubleplus-** + cold → **doublepluscold**(≡ superlatively cold)
- **Suffixes:** Formed adjectives or adverb
 - **Adjective:** speed + **-ful** → speedful(≡ rapid)
 - **Adverb:** speed + **-wise** → speedwise(≡ rapidly)

5. Newspeak

The A Vocabulary: Everyday Life Uses

- **Abbreviations:** Made the meaning exactly **opposite** or **ambiguous**
 - joycamp(= forced-labour camp)
 - Minipax(= Ministry of **Peace** = Ministry of **War**)
 - Minitrue(= Ministry of **Truth** = Ministry **manipulating** historical records)
 - Miniluv(Ministry of **Love** = Ministry **forcing** rebels to love Big Brother by tortures)
- **Inflections:** No more irregularities and analogy totally works
 - **Verb:** speak - spoke - spoken → speak - **speaked** - **speaked**
 - **Adjective:** good - better - best → good - **gooder** - **goodest**
 - ex. pronouns, relatives, demonstrative adjectives and axillary verbs

5. Newspeak

The B Vocabulary: Political Uses

- **Vocabulary:** Consists of compounds
 - good + think → **goodthink**(≡ orthodoxy)
 - **Verb:** goodthink - goodthinked - goodthinked(≡ to think something in an orthodox way)
 - **Adjective:** goodthinkful(≡ orthodox)
 - **Adverb:** goodthinkwise(≡ orthodoxically)
- **Inflections:** Closer to Oldspeak or Standard English than The Vocabulary A
 - Minitrue > Minitruthful: *-trueful* is awkward
 - Minipax > Minipeaceful: *-paxful* is awkward
 - Miniluv > Minilovely: *-lovful* is awkward

5. Newspeak

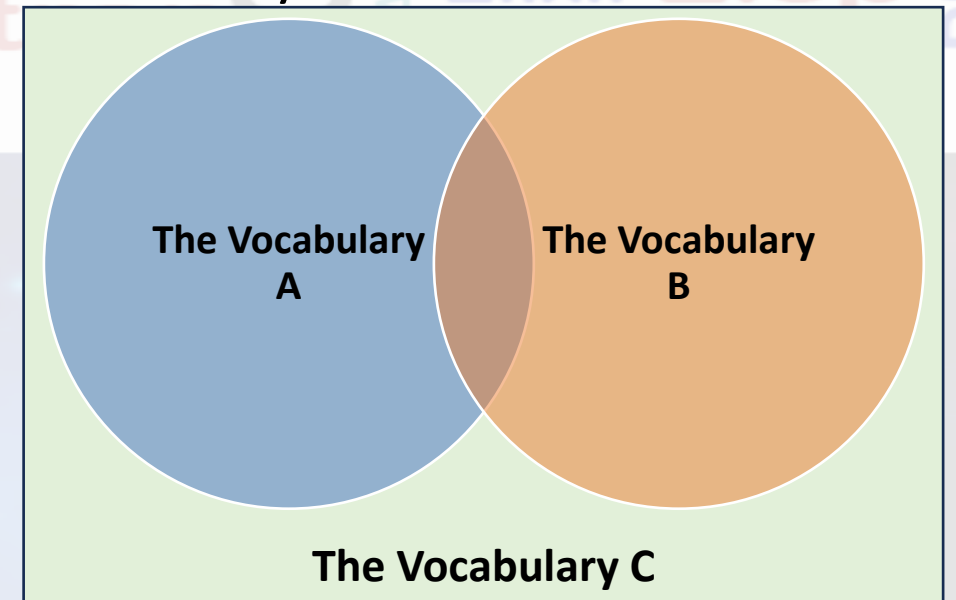
The B Vocabulary: Political Uses

- **Meaning of words:** Required to master Newspeak to use and/or understand
 - In the following sentences, each corresponding part is highlighted in the same colour
 - **Oldspeak:** *Those whose ideas were formed before the Revolution cannot have a full emotional understanding of the principle of English Socialism.*
 - **Newspeak:** *Oldthinkers unbellyfeel Ingsoc.*
 - As you see, each word has too ambiguous meaning to understand at a glance
- **Word purging:** Undesirable meanings in each word were exterminated
 - e.g. *free* means a status where something was removed as shown *be free from ~*. No more *liberty* nor something related to anti-governmental concepts
- **Duckspeak:** The concept that an act was evaluated totally based on its behaviour

5. Newspeak

The C Vocabulary: Scientific and Technical Uses

- **Role:** Supplementary element of The Vocabulary A and The Vocabulary B
- **Word purging:** Undesirable meanings in each word were exterminated
- **Grammar:** Same as The Vocabulary A and The Vocabulary B
- **Use case:** Highly limited to science.
- **Irony:** The term science itself was **absent**
 - The word [Ingsoc](#) covered it



6. *Why Nineteen Eighty-Four?*

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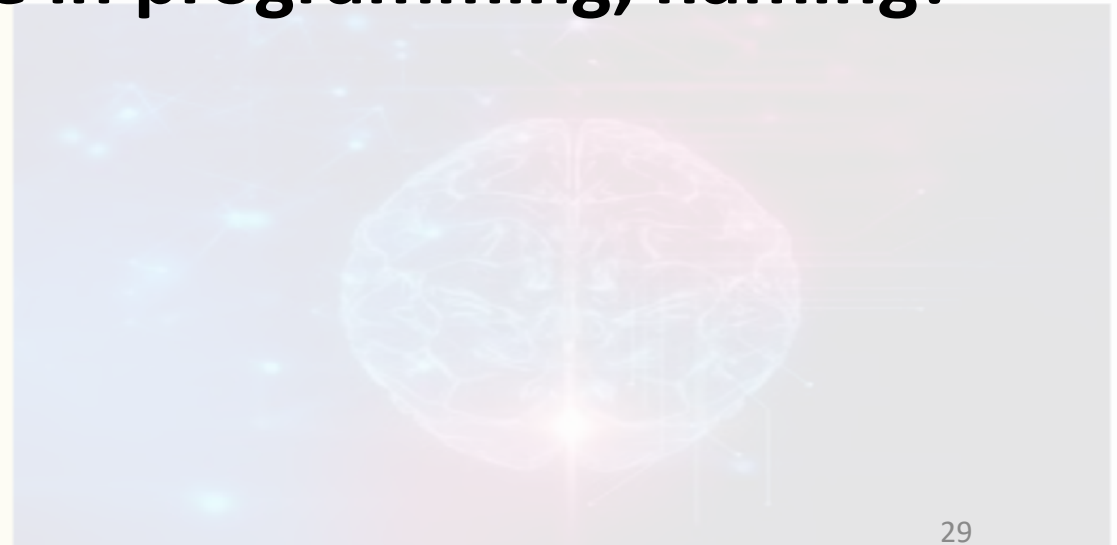
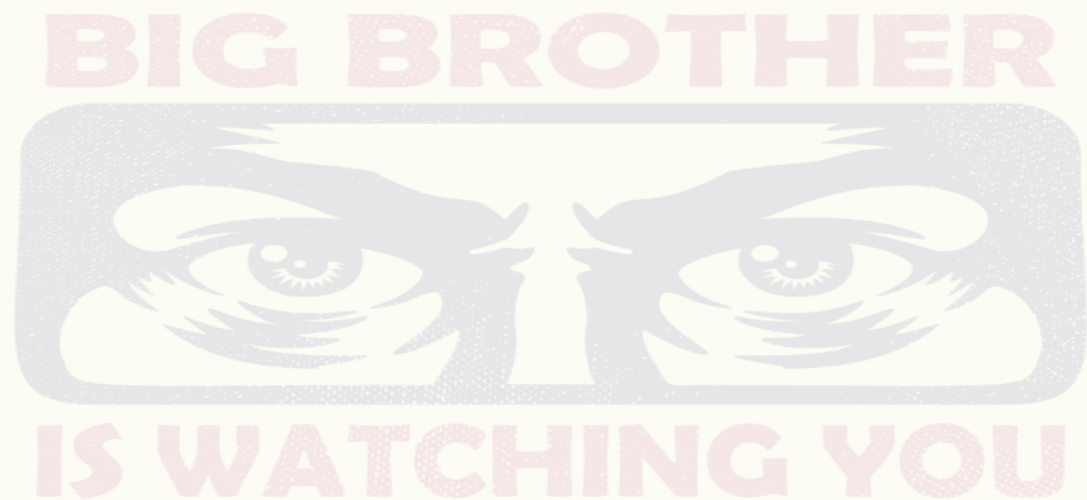
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6. Why *Nineteen Eighty-Four*?

To cut a long story short, *Nineteen Eighty-Four* gives us a deep insight into...

Our developers' forever challenge in programming, naming!



6. Why *Nineteen Eighty-Four*?

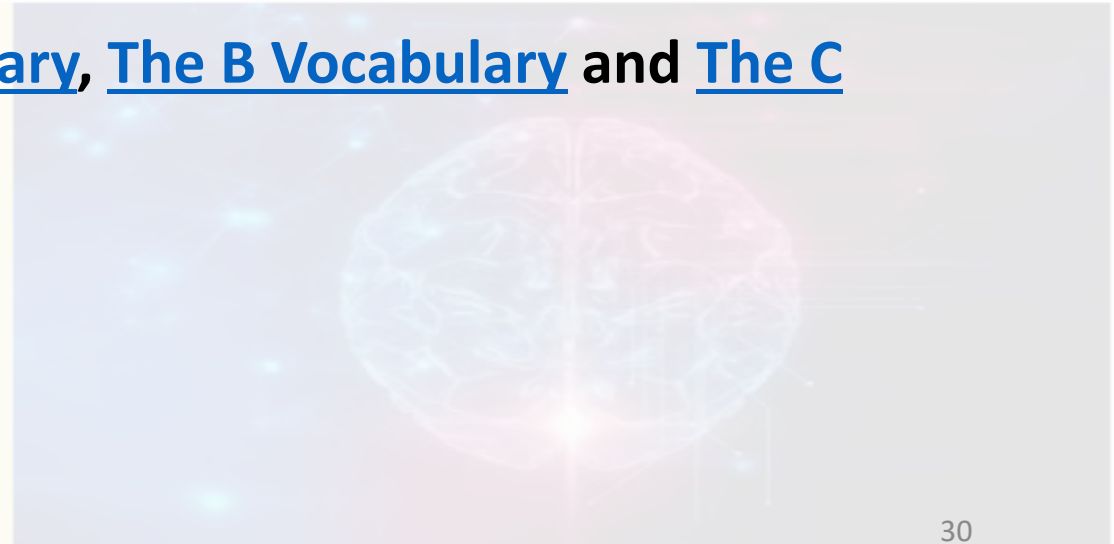
We know naming is also a difficult work due to its trade-offs.

The longer the name is, the more redundant it is.

The shorter the name is, the more possible it fails to embody the important factors.

How should we name something in a **concise** and **accurate** way?

The answer is **word forming** of [The A Vocabulary](#), [The B Vocabulary](#) and [The C Vocabulary](#) in the previous chapter.



6. Why *Nineteen Eighty-Four*?

The words below emerged on social media and have got citizenship as you know. They represent addition of prefix *un-* to existing words in [The A Vocabulary](#).

- friend ↔ **unfriend**
- like ↔ **unlike**
- star ↔ **unstar**
- flag ↔ **unflag**
- follow ↔ **unfollow**



6. Why *Nineteen Eighty-Four*?

Naming should be the last thing AI is good at the first thing only we human developers must pay attention to.

The creator of Ruby Mr. Yukihiro Matsumoto, generally called **Matz**, is known as a strict developer on naming, according to a popular book [107. The Importance of Naming on Japanese Edition of 97 Things Every Programmer Should Know](#).

That is because **he believes naming reflects the system architecture**.

Only human developers can take it into account in naming of a class, module, method and so on, as of today.

* For the details of an interesting episode, refer to [Road to Ruby for A Linguistics Nerd > 3. The Most Memorable Discussion Mr. Matz](#).

6. *Why Nineteen Eighty-Four?*

Here is the very reason we developers use high-level programming languages in this highly advanced AI era.

What is it?

It is because **they are the only means by which we can express how we design, recognise and implement the architecture.**

Most codes are written and read by human developers today, so they must be as 'self-evident' as possible by appropriate naming at a glance.

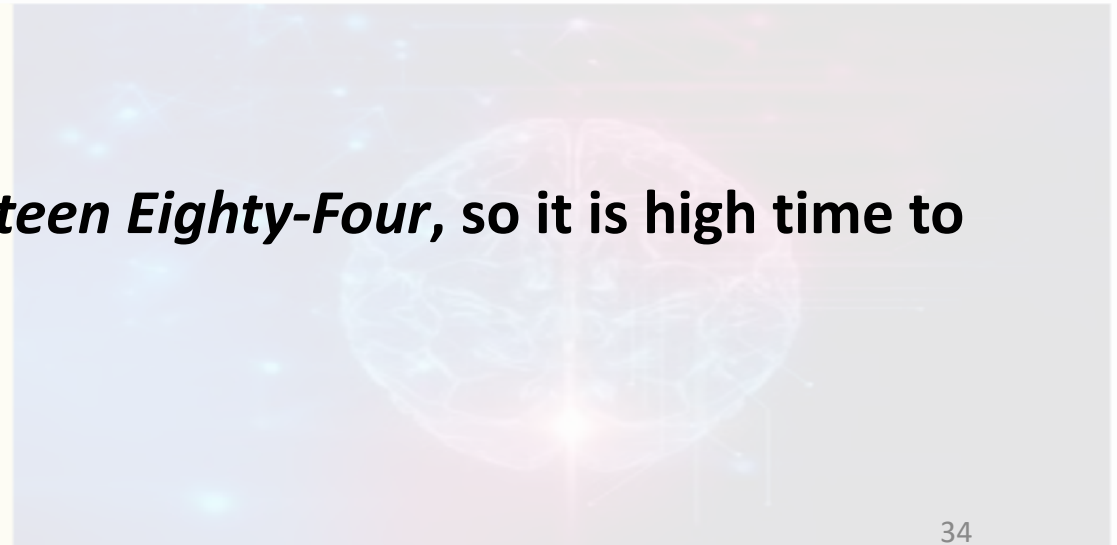
You will learn universal linguistic principles helpful for naming from Newspeak!

6. Why *Nineteen Eighty-Four*?

George Orwell, the author of *Nineteen Eighty-Four*, is highly likely to have predicted the following present-day affairs as we see today while he was writing it.

- New types of word formation on social media
- Political correctness and censorship
- Ongoing dictatorship and wars among some countries
- etc.

You will learn something universal from *Nineteen Eighty-Four*, so it is high time to read it!

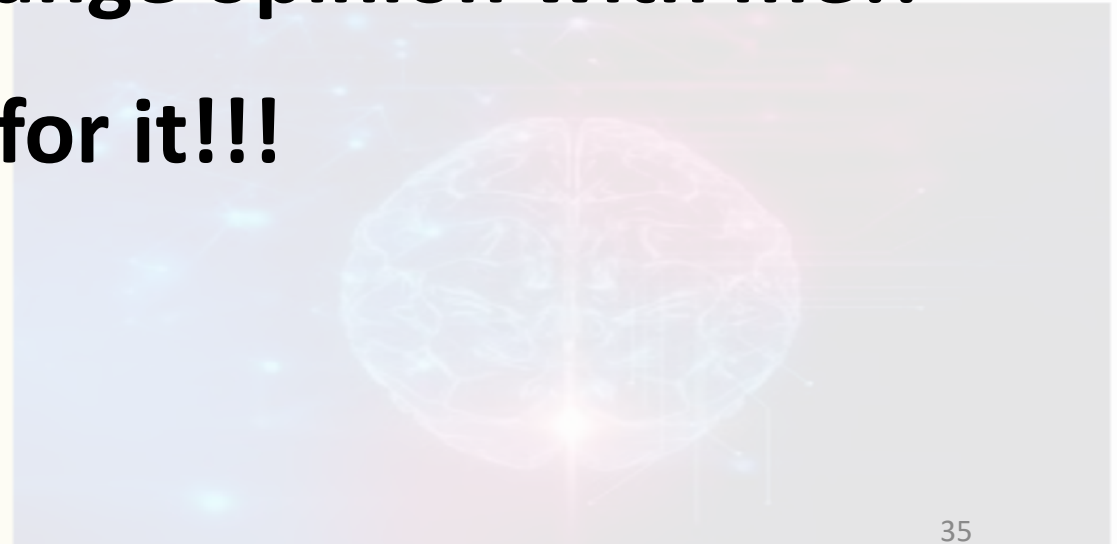


6. Why *Nineteen Eighty-Four*?

There is no reason not to read *Nineteen Eighty-Four*!

Share your reviews and exchange opinion with me!!

I am excited for it!!!



7. References

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- Hayato Ishida, [*Road to Ruby for A Linguistics Nerd*](#), Tokyo, Hayato Ishida, 2025

